

#01 Predictive Maintenance Alert System

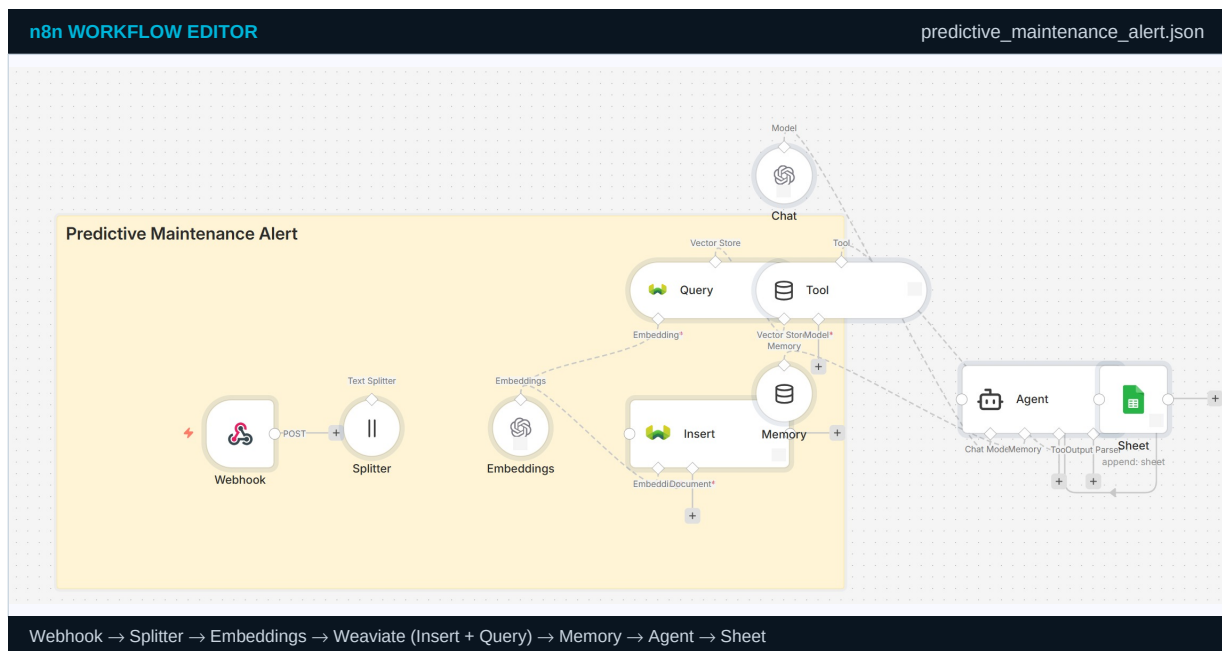
IoT / Industry 4.0

RAG-powered sensor analysis — detect failure signatures before breakdown occurs

BUSINESS PROBLEM

Unplanned equipment failure causes production losses averaging \$260,000 per hour in heavy manufacturing. Traditional threshold-based alarms trigger too late or too often. This workflow ingests raw sensor telemetry every 15 minutes, semantically matches it against a library of historical failure signatures using vector search, and predicts failure windows with confidence scores — giving maintenance teams 4-8 hours advance notice to act before breakdown occurs.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Webhook POST — sensor telemetry (MQTT gateway → n8n, every 15 min)

OUTPUT — n8n output item — written to Google Sheets + Slack alert fired

```
{
  "machine_id": "CNC-Mill-07",
  "timestamp": "2024-06-15T02:14:33Z",
  "sensors": {
    "temp_bearing_c": 94.7,
    "vibration_rms": 11.8,
    "current_amp": 18.4,
    "oil_pressure_bar": 2.1,
    "rpm": 2840
  },
  "shift": "night",
  "site": "Factory-Site-A"
}
```

```
{
  "json": {
    "machine_id": "CNC-Mill-07",
    "assessed_at": "2024-06-15T02:14:41Z",
    "risk_score": 0.91,
    "risk_level": "CRITICAL",
    "predicted_failure_window": "3-5 hours",
    "matched_pattern": {
      "label": "Spindle bearing fatigue",
      "similarity": 0.92,
      "ref_date": "2023-09-14"
    },
    "flags": {
      "temp_bearing_c": "ABOVE LIMIT (max 90 C)",
      "vibration_rms": "CRITICAL (max 10 RMS)",
      "trend": "+2.3 RMS rising over 4h"
    },
    "action": "Replace spindle bearing before 05:00",
    "parts": ["SKF-6205-2RS"],
    "ticket": "MAINT-4821"
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Webhook — POST from MQTT gateway	IoT gateway sends sensor batch every 15 min. Payload: machine_id, all sensor readings, shift and site. HTTP 200 returned immediately to gateway.
2	✂ Text Splitter — chunk maintenance knowledge	Historical failure reports and machine manuals split into 500-char chunks with 100-char overlap. Machine type prepended to each chunk as context prefix before embedding.
3	■ Embeddings — OpenAI text-embedding-ada-002	Each chunk → 1536-dim vector. Batch: 20 chunks per API call. Metadata tagged per chunk: {machine_type, failure_mode, severity, incident_date}.
4	■ Weaviate Insert — persist readings + patterns	Current sensor reading stored. 90-day rolling window maintained. Schema: {machine_id, sensor_values, failure_mode, risk_level}.
5	■ Weaviate Query — semantic failure match	Incoming sensor vector queried against failure library using cosine similarity. Returns top-3 most similar historical incidents with similarity scores.
6	■ Vector Store Tool — exposes retriever to Agent	Wraps vector retriever as a callable tool. Agent invokes this during reasoning loop to fetch matched failure descriptions, RCA documents, and recommended actions.
7	■ Window Buffer Memory — 10-reading window	Stores last 10 readings (2.5h coverage). Agent uses this to detect trends: is vibration rising? Is temperature rate-of-change accelerating beyond baseline?
8	■ OpenAI GPT-4 — risk reasoning	Receives: matched patterns + current readings + trend data. Outputs structured risk assessment. Temperature 0.1 for deterministic, consistent scoring.

#	WHAT HAPPENS	PRACTICAL DETAIL
9	■ AI Agent — orchestration loop	Iteratively calls Vector Tool → Memory → Chat Model. Produces: {risk_score 0-1, risk_level, predicted_failure_window, matched_pattern, action, parts_required}.
10	■ Google Sheets — append log row	Appends: timestamp, machine_id, risk_score, risk_level, predicted_window, matched_pattern, action_taken, parts_required, ticket_id. Feeds maintenance KPI dashboard.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
OpenAI API Key	platform.openai.com/api-keys	OpenAI API
Weaviate Cloud URL + Key	console.weaviate.cloud → Create cluster → API Keys	Weaviate API
Google Sheets OAuth2	console.cloud.google.com → Sheets API → OAuth credentials	Google Sheets OAuth2
Slack Bot Token	api.slack.com/apps → OAuth & Permissions → Bot Token: chat:write	Slack API

SAMPLE RUN — what this workflow processed and produced

Scenario	A food packaging plant runs 12 conveyor belt motors 24/7. Each motor posts vibration, temperature, and current readings every 15 minutes via ESP32 sensor nodes connected to an MQTT broker.	
Input received	Motor CNC-Mill-07 posted: vibration_rms 11.8 (baseline 7.2), bearing temp 94.7°C (baseline 71°C), current draw 18.4A (rated 16A). Night shift did not flag it — within old threshold limits.	
What the workflow found	Weaviate query matched reading profile against incident #INV-2021-039: 'bearing overheating + vibration spike, failed 6h after similar reading.' Similarity score: 0.92.	
Output produced	Risk score 0.91 CRITICAL. Predicted window: 3-5 hours. Slack alert sent at 02:14. Ticket MAINT-4821 created. Bearing replaced at 04:30 — motor back in service by 05:00, zero line downtime.	
JSON: predictive_maintenance_alert.json	GitHub: trm-predictive-maintenance-agent	n8n.io/workflows/predictive-maintenance/